

Demographic Vote Sensitivity Analysis — Tester Guide

Live app: <https://demographic-voteanalysis.vercel.app>

A plain-English guide to what this tool does, every feature it has, and — for each one — exactly how to try it yourself and what the result means.

The one-sentence version

This tool builds each congressional district's **eligible electorate from real Census data** — how many citizens of voting age live there, split by race and age — lets you **turn dials** to see how the projected Democratic vote share moves, and then lets you **check that projection against what actually happened** in five real elections. It covers all 28 districts in Georgia and North Carolina.

What it is — and what it is NOT

It IS a *what-if /* sensitivity tool. It answers: "Given who lives in this district and a set of assumptions about turnout and party preference, what does the math imply?" Every assumption is visible and adjustable.

It is NOT, by default, a prediction — the "2026 Projection" mode is a hypothetical built from your slider settings, not a forecast. (Like a mortgage calculator: it doesn't predict whether you'll buy the house, it shows what the numbers do when you change the inputs.)

But it also shows you real results. Four of the five modes display **actual historical election outcomes** — not model output. The banner color at the top of the page always tells you which kind of number you're looking at:

Banner color	Mode	What you're seeing
Blue	2026 Projection	Hypothetical. Built from your sliders. Not a real result.
Amber	Validate: 2022 House, 2020 President, 2020/2022 Senate runoffs	Real. Actual historical election results, sourced and cited.

Where the numbers come from

- **The people:** U.S. Census Bureau Citizen Voting-Age Population (CVAP) — citizens 18+. Non-citizens are excluded automatically.
- **The behavior (slider defaults):** national turnout rates (Census CPS survey) and national vote-preference splits (AP VoteCast) per race/age group — defaults you're meant to override, not facts.
- **The validation data (real results):** actual, cited election returns for five races, cross-checked before shipping.

Everything traces to a public source and is cached locally.

Every feature, and how to test it yourself

1. State picker & sliders

Pick Georgia or North Carolina, then drag a slider for any racial group's **turnout** or **Dem share**. The slider label shows the real, current percentage and updates live.

Try it: note "Statewide GA: 55.9% Democratic" at the top. Drag "White — % voting Dem" down to ~25% (roughly the Deep South rate). Watch the statewide number drop and several districts flip from D to R.

2. Preset scenarios

One-click buttons (`base`, `southern_white`, `high_black_turnout`, `youth_surge`, `hispanic_shift_r`) set several sliders at once to a named, documented combination.

Try it: click `southern_white`. The seat count should drop from "11 of 14" to "**5 of 14**" — the actual real-world Georgia Democratic seats.

3. The map

An SVG map colored by margin (red = more Republican, blue = more Democratic). Recolors live as you adjust sliders or switch validation modes.

Try it: click district "7." The district detail, county legend, and flip calculator on the right update to that district.

4. County legend

For the selected district, every county in its jurisdiction. Counties split across multiple districts are flagged "(part)".

Try it: select GA-07 — it lists "Gwinnett (part)" and "Fulton (part)," both correctly flagged as shared with neighboring districts.

5. District detail (race × age electorate)

The full citizen-voting-age-population table broken down by race and age band, with row/column totals — the raw demographic material the model is built from.

6. Flip calculator (tipping point)

Shows how far **one single change** — e.g. "White — Dem share" — would have to move, on its own, to bring the district to 50/50 (projection mode) or to match a real historical result (validation mode).

Try it: in 2026 Projection mode, select GA-07 — White Dem share needs to move about **-19 points** to reach 50%. Switch to "Validate: 2022 House" — the target changes to GA-07's real 2022 result, and the same lever needs a much smaller move.

7. Save / load scenarios (your browser only)

Name your current slider settings, save, reload, or delete later — stored in your browser's local storage.

8. Validate: 2022 House

The actual November 2022 U.S. House result, district by district, for both Georgia and North Carolina. The table gains Actual / Your model / Diff / Winner columns.

Try it: the seatbar should read "5 of 14 seats D" for Georgia and "7 of 14 seats D" for North Carolina — real, certified outcomes.

9. Validate: 2020 President

The actual 2020 presidential vote, re-tabulated onto the same district lines this model uses. **Georgia only** (see limitations below for why North Carolina isn't included).

Try it: with Georgia selected, seatbar reads "6 of 14 seats D." Switch to North Carolina — you'll see an explicit message explaining why it isn't available.

10. Validate: 2020 Senate runoffs (GA)

A candidate dropdown lets you pick Jon Ossoff (vs. Perdue) or Raphael Warnock (vs. Loeffler) — both decided in the January 5, 2021 runoff. **Georgia only.**

Try it: defaults to Ossoff — seatbar reads "6 of 14 seats D." Switch to Warnock — very similar numbers. Switch state to North Carolina — the message explains NC doesn't hold Senate runoffs at all.

11. Validate: 2022 Senate runoff (GA)

Raphael Warnock vs. Herschel Walker, December 6, 2022. **Georgia only.**

Try it: seatbar reads "5 of 14 seats D." Select GA-07 — Warnock runs well ahead of the 2020 baseline there, a well-known real pattern.

A worked example — "Can Democrats hold Georgia's 7th if young voters stay home?"

1. **Explore.** Pick Georgia, click GA-07. About 467,000 voting-age citizens — 39% White, 33% Black, 13% Hispanic, 12% Asian. The most diverse district in the state.
2. **See the honest starting point.** Default sliders show GA-07 at ~59% Democratic — national numbers run too Democratic in the South.
3. **Make it realistic.** Click `southern_white`. GA-07 slides to ~50.5% — a Tossup — and the seat counter drops to "5 of 14."
4. **Check it against reality.** Switch to "Validate: 2022 House." GA-07's actual 2022 result was 61.1% Democratic — even stronger than the calibrated projection.
5. **Ask your what-if.** Back in Projection mode with `southern_white` applied, drag Black and Hispanic turnout down to ~85%. GA-07 tips just below 50%.
6. **Get the exact number.** The flip calculator tells you precisely how far Black turnout alone would have to fall to make GA-07 a 50/50 race.

Honest limitations (worth knowing)

- **National defaults need local tuning** — Southern rural White voters are much more Republican than the national average; the sliders and `southern_white` preset correct for this.
- **Race-by-age is estimated, not measured** — Census gives race and age splits separately, not the exact combination; the tool assumes a district's age mix is similar across races.
- **District lines are the 118th-Congress (2022-election) map.** Both Georgia and North Carolina redrew again in 2023 — confirm the enacted map before using this for a live 2026 cycle.
- **NC 2020 President is not shown, on purpose** — the only source found was actually on NC's old, 13-district map. Rather than show a mismatched number, the app says so honestly.
- **The 2021 Senate runoffs used Georgia's previous map** — shown anyway, backed by a documented cross-check that the two maps are close enough for the comparison to be meaningful.
- **Turnout and preference are assumptions in projection mode** — that's the point: you own them, the tool makes them explicit instead of hiding them.

In short

Load the real electorate → adjust the assumptions → watch the outcome move → check it against real results → get exact numbers on what it would take to change the outcome. Five real elections are built in specifically so you never have to take the model on faith.